

Math 290

Practice Quiz 2 — Version A Solutions

1(a) Answer: B. The theorem is false: take $a = 2$, $b = 1$. Then $2 \mid 2 = 2b$, but $2 \nmid 1$. (The error in the proof is the claim $c/2 \in \mathbb{Z}$; nothing forces c to be even.) (Based on Proposition 7.12.)

1(b) Answer: True. The premise $x^2 < 0$ is never true for $x \in \mathbb{R}$, so the implication is vacuously true. (Based on Example 6.15.)

1(c) Answer: False. $2 - (-2) = 4$ and $3 \nmid 4$, so $2 \not\equiv -2 \pmod{3}$. (Based on Example 8.12.)

2. (Based on Exercise 7.2.)

Solution. Let $x, y \in \mathbb{Z}$. We work contrapositively. Assume x is even and y is odd. Then $x = 2k$ and $y = 2\ell + 1$ for some $k, \ell \in \mathbb{Z}$. We find

$$xy + y^2 = y(x + y) = (2\ell + 1)(2k + 2\ell + 1) = 4k\ell + 4\ell^2 + 4\ell + 2k + 1 = 2(2k\ell + 2\ell^2 + 2\ell + k) + 1.$$

Since $2k\ell + 2\ell^2 + 2\ell + k \in \mathbb{Z}$, the number $xy + y^2$ is odd. $\square$

3. (Based on Exercise 9.4 and Theorem 9.8.)

Solution. Lemma: if $a \in \mathbb{Z}$ and $3 \mid a^2$, then $3 \mid a$. We work contrapositively. Assume $3 \nmid a$. Then $a = 3k + 1$ or $a = 3k + 2$ for some $k \in \mathbb{Z}$. If $a = 3k + 1$, then $a^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1$. If $a = 3k + 2$, then $a^2 = 9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1$. In both cases a^2 has the form $3m + 1$ with $m \in \mathbb{Z}$, so a^2 is not of the form $3m$ (every integer is of exactly one of the forms $3m$, $3m + 1$, $3m + 2$); hence $3 \nmid a^2$.

Now assume, by way of contradiction, that $\sqrt{3} \in \mathbb{Q}$. Write $\sqrt{3} = a/b$ with $a, b \in \mathbb{N}$ and a/b in lowest terms. Squaring and clearing denominators gives $a^2 = 3b^2$, so $3 \mid a^2$, and by the lemma $3 \mid a$. Write $a = 3m$ with $m \in \mathbb{Z}$. Then $9m^2 = 3b^2$, so $b^2 = 3m^2$. Thus $3 \mid b^2$, and so $3 \mid b$. Now 3 divides both a and b , contradicting that a/b is in lowest terms. Hence $\sqrt{3}$ is irrational. $\square$

4. (Based on Exercise 10.5.)

Solution. ($\subseteq$) Let $x \in X - (X - Y)$. Then $x \in X$ and $x \notin X - Y$ (definition of set difference). Since $x \notin X - Y$, we have $\neg(x \in X \wedge x \notin Y)$ (definition of set difference). So $\neg(x \in X)$ or $\neg(x \notin Y)$ (De Morgan's law). That is, $x \notin X$ or $x \in Y$ (double negation). Since $x \in X$, the first option is false, so $x \in Y$ (disjunctive syllogism: $(P \vee Q) \wedge \neg P \Rightarrow Q$). Thus $x \in X$ and $x \in Y$, so $x \in X \cap Y$ (definition of intersection). Therefore $X - (X - Y) \subseteq X \cap Y$ (definition of subset).

($\supseteq$) Let $x \in X \cap Y$. Then $x \in X$ and $x \in Y$ (definition of intersection). Since $x \in Y$, we have $x \notin X - Y$ (tautology $Q \Rightarrow P \vee Q$). So $\neg(x \in X)$ or $\neg(x \notin Y)$ (double negation). Hence $\neg(x \in X \wedge x \notin Y)$ (De Morgan's law), i.e. $x \notin X - Y$ (definition of set difference). Since $x \in X$ and $x \notin X - Y$, we get $x \in X - (X - Y)$ (definition of set difference). Therefore $X \cap Y \subseteq X - (X - Y)$ (definition of subset).

Both inclusions hold, so $X - (X - Y) = X \cap Y$ (definition of set equality). $\square$